📘 MODULE 18: CONFIGURATION FILES

18.1 Concept

Splunk's behavior is controlled by plain-text `.conf` **files** made of **stanzas** (`[name]`) and **settings** (`key = value`). The UI just edits these files for you. Admins and interviewers expect you to know them.

```ini
[stanza_name]
setting1 = value
setting2 = value
```

18.2 Rules

- Use `local/` for changes, and never edit `default/`.
- Changes to **index-time** settings (inputs, props parsing, transforms for routing) need a **restart** on the instance that does the parsing. Many search-time settings only need a refresh.
- Settings are **merged by attribute**, not file replaced (Module 19).
- A typo in a stanza name or setting is **silently ignored**, so verify with `btool`.

18.3 The ten files

1. `inputs.conf` (what to collect)

Location: where the data is collected (UF, HF, or standalone), usually `etc/system/local/` or an app's `local/`.

```ini
[monitor:///var/log/auth.log]
index = linux_auth
sourcetype = linux_secure
disabled = 0
```

Common mistake: wrong index name (data is dropped), or a Splunk user without read permission.

2. `outputs.conf` (where to send)

Location: on the forwarder.

```ini
```

```
[tcpout]
defaultGroup = my_indexers

[tcpout:my_indexers]
server = 192.168.1.10:9997
```

`defaultGroup` names the target group, and `server` is `host:port` of the indexer's receiving port.
Mistake: the receiver isn't enabled, or the port is blocked.

3. `props.conf` (how to parse, per sourcetype/source/host)

ini

```
[my_app_logs]
SHOULD_LINEMERGE = false
TIME_FORMAT = %Y-%m-%d %H:%M:%S
TIME_PREFIX = ^
EXTRACT-user = user=(?<user>\w+)
```

It controls line breaking, timestamps, extractions, and calls to transforms. **Mistake:** index-time settings placed on the wrong tier (they must be on the first full Splunk instance the data touches: indexer or HF, **not** the UF for most parsing).

4. `transforms.conf` (reusable regex-based operations)

ini

```
[drop_debug]
REGEX = \bDEBUG\b
DEST_KEY = queue
FORMAT = nullQueue
```

It is called from props: `TRANSFORMS-drop = drop_debug`. Covered deeply in Module 23.

5. `indexes.conf` (index definitions)

ini

```
[linux_auth]
homePath   = $SPLUNK_DB/linux_auth/db
coldPath   = $SPLUNK_DB/linux_auth/colddb
thawedPath = $SPLUNK_DB/linux_auth/thaweddb
frozenTimePeriodInSecs = 7776000
```

Mistake: forgetting `thawedPath`, or not restarting.

6. `server.conf` (instance identity and core behavior)

```ini
[general]
serverName = splunk-lab
```

It also holds clustering, SSL, and license-manager pointer settings. **Mistake:** two instances with the same `serverName`, which confuses monitoring.

7. `limits.conf` (resource limits)

```ini
[subsearch]
maxout = 10000
maxtime = 60

[searchresults]
maxresultrows = 50000
```

This is where the subsearch and `join` limits from Module 12 and 13 live. **Mistake:** raising limits instead of fixing a poor search.

8. `authentication.conf` (how users log in)

```ini
[authentication]
authType = LDAP
authSettings = corp_ldap

[corp_ldap]
host = ldap.example.com
port = 636
SSLEnabled = 1
```

`authType` can be `Splunk` (built-in), `LDAP`, or `SAML`/scripted. SAML has its own stanzas. Normally configured in the UI.

9. `authorize.conf` (roles and capabilities)

```ini
```

```
[role_soc_analyst]
importRoles = user
srchIndexesAllowed = linux_auth;windows;main
srchIndexesDefault = linux_auth
srchJobsQuota = 5
schedule_search = enabled
```

It defines **what a role can search and do**. Note `srchIndexesAllowed` uses `;` separators.

10. `web.conf` (Splunk Web)

ini

```
[settings]
httpport = 8000
enableSplunkWebSSL = true
```

It controls the web port, SSL, and timeouts.

18.4 Summary table

File	Controls	Lives mostly on
inputs.conf	What data to collect	UF / HF / indexer
outputs.conf	Where to forward	UF / HF
props.conf	Parsing + extraction per sourcetype	Indexer/HF (index-time), search head (search-time)
transforms.conf	Regex operations (route, mask, extract)	Same as props
indexes.conf	Index definitions	Indexer
server.conf	Instance identity, clustering	All
limits.conf	Resource limits	Search head / indexer
authentication.conf	Login method	Search head
authorize.conf	Roles	Search head
web.conf	Splunk Web	Search head

18.5 Inspect with btool

bash

```
sudo /opt/splunk/bin/splunk btool inputs list monitor --debug
```

- `btool` shows the **merged, effective** configuration.
- `inputs list monitor` lists the `monitor` stanzas of `inputs.conf`.
- `--debug` shows **which file** each line came from. This is your best configuration troubleshooting tool.
- **Expected output:** each setting prefixed by its file path.
- **Errors:** "command not found" means use the full path. Empty output means the stanza isn't loaded (typo, wrong folder).

Check syntax: `sudo /opt/splunk/bin/splunk btool check`.

18.6 Hands-on

1. Open `/opt/splunk/etc/system/default/` and list the `.conf` files.
2. Find your own input: `sudo /opt/splunk/bin/splunk btool inputs list monitor --debug`.
3. Create a role `soc_analyst` in the UI (Settings → Roles), then find it in `etc/system/local/authorize.conf` (or the app you chose).
4. Check what limits apply: `splunk btool limits list subsearch --debug`.

18.7 Interview Q&A

- **inputs vs outputs?** inputs = what to collect, outputs = where to send.
- **Where do you assign sourcetype?** `inputs.conf` (at collection) or `props.conf` rules (by source/host).
- **props vs transforms?** props attaches settings to a sourcetype, transforms holds reusable regex rules that props calls.
- **How do you see the effective config?** `splunk btool <file> list --debug`.
- **Where should config changes go?** `local/`, never `default/`.
- **Which file defines index retention?** `indexes.conf`.

18.8 Revision

Stanzas + key=value · edit `local/` only · inputs/outputs/props/transforms/indexes/server/limits/authentication/authorize/web · `btool --debug` · typos are silent.